

Assignment 1 - Web Development

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Frontend development focuses on the part of a website that users see and interact with. It uses HTML, CSS, and JavaScript. Example: the Amazon homepage, search bar, and product pages.

Backend development handles server-side logic, authentication, business rules, and database communication using technologies like Java, Python, Node.js, or PHP. Example: processing Amazon orders and verifying user login.

Full-stack development combines both frontend and backend skills. A full-stack developer can build an entire web application from the user interface to the database.

2. Create a simple diagram showing how the client-server model works in web architecture.

Client (Browser) → HTTP Request → Web Server → Database (if needed) → Web Server → HTTP Response → Client (Displays Web Page)

3. Describe how a browser requests and displays a web page from a web server.

The browser sends an HTTP request to the web server. The server processes the request, retrieves data if required, and sends HTML, CSS, JavaScript, and other resources back to the browser. The browser then renders these resources and displays the webpage.

4. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

VS Code: code editor. Google Chrome: testing websites. Chrome DevTools: debugging. Git: version control. GitHub: code hosting. Node.js: JavaScript runtime. npm: package manager. Live Server: automatically refreshes the browser.

5. Explain what a web server is and give examples of commonly used servers.

A web server receives requests from web browsers and sends webpages or data back to users. Common examples are Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed, and Apache Tomcat.

6. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Frontend Developer: builds the user interface. Backend Developer: develops server-side logic and APIs. Database Administrator (DBA): designs, secures, backs up, and maintains databases.

7. Install VS Code and configure it for HTML, CSS, and JavaScript development. Take a screenshot of the setup.

Install VS Code, install the Live Server, HTML CSS Support, JavaScript Snippets, and Prettier extensions. Create index.html, style.css, and script.js files. Insert your own screenshot of the configured setup.

8. Explain the difference between static and dynamic websites. Provide an example of each.

A static website displays the same content to every visitor (example: personal portfolio). A dynamic website changes content based on users or database information (example: Amazon, Facebook).

9. Research and list five web browsers. Explain how rendering engines differ between them.

Google Chrome (Blink), Microsoft Edge (Blink), Mozilla Firefox (Gecko), Apple Safari (WebKit), Opera (Blink). Rendering engines interpret HTML, CSS, and JavaScript to display webpages. They differ in speed, optimization, and standards implementation.

10. Draw a labeled diagram showing the basic web architecture flow — client, server, database, and APIs.

Client → Web Server → Database / APIs → Web Server → Client